Review of P14 — MS1

1 Summary

AuroraCast forecasts the planetary Kp index, meaning how disturbed Earth's magnetic field is from 0 to 9. This happens for the next 3-hour interval from live NOAA solar-wind data and the probability of a storm when Kp ≥ 5. The forecast is made at each interval start and scored against the published Kp of GFZ with persistence (next value = last value) as the baseline. It is mainly for aurora watchers but also for HF-radio and GNSS users.

2 Strengths

The success criterion is tiered (minimum, target, stretch) and covers MAE and storm recall at a stated precision, with a live rolling 30-day MAE as a final check. The proposal also says the targets will be rejustified if persistence proves strong, which seems fitting towards a baseline that has not been measured yet.

The leakage section names four concrete risks instead of only a chronological split: windows end before t, the GFZ publication delay for lagged Kp, nowcast versus definitive Kp and the L1-to-Earth time shift of the OMNI history. Each comes with a suggested fix and one shared feature module for training and serving is a good answer to the last one.

The data section is quantified and the 82'000 labelled intervals seems right for the 28 years at 8 intervals per day. The label comes from ground magnetometers, so it is physically independent of the satellite inputs.

3 Concerns or gaps

The persistence baseline is the last Kp published before t. If the nowcast of GFZ arrives with a delay, that baseline looks back more than one interval and is weaker than ordinary persistence, meaning "10% below persistence" may be cheap. NOAA SWPC's own forecast is the higher standard, but it is only logged as an external reference and has no threshold.

The 2024 and 2025 test years use OMNI data, which is quality-controlled, while the live system consumes the noisier real-time feed. The train/serve difference is called small and known, but no number is given and the offline test score will not show it. Only the live 30-day MAE will.

The stack (Airflow, TimescaleDB, MLflow, Streamlit, Docker Compose on a VM) is heavy for a solo core pipeline, as listed in the Proposal Guide as a common mistake.

4 One concrete suggestion

Run the EDA the proposal already defers and compute persistence two ways: last published Kp and last completed interval. Pin the 10% target to the stronger one. Report the storm base rate in the same pass and give the SWPC forecast an explicit target for the live period.